

Prediction Of Churn and Credit Decisions Using Artificial Intelligence Advanced Classifier

Thamaraikannan N S^a, Prabakaran J^{b*}

^a Department of Computing Technologies, School of Computing, SRM Institute of Science and Technology, Kattankulathur - 603203, India.
ts1262@srmist.edu.in

^b Department of Networking and Communications, Faculty of Engineering and Technology, SRM Institute of Science and Technology, Kattankulathur - 603203, India. prabakaj@srmist.edu.in

Abstract: Machine learning plays a key role in the financial sector, enabling customer retention and data-driven loan decision-making. This study introduces an AI-powered model for predicting bank churn and loan eligibility using a stacking ensemble classifier. The research utilizes two datasets, Churn Data and Credit Eligibility, which contain customer records related to churn behaviour and loan approvals. To address class imbalance, RandomOverSampler was applied, ensuring balanced representation for accurate predictions. The model incorporates multiple classifiers, including Voting Classifier, Extra Trees Classifier for customer retention and part of Stacking Classifier, Random Forest, Support Vector Machine (SVM), Decision Tree and Logistic Regression, with Logistic Regression serving as the meta-classifier in order to make right credit decisions. Trained using five-fold cross-validation, the model achieved an accuracy of 97%, demonstrating its robustness in predictive analysis. After evaluation, the trained models are deployed, allowing bankers to review customer retention insights and directly assess loan eligibility by the banker or customer. By leveraging ensemble learning, the system enhances decision accuracy, helping financial institutions identify customers at risk of attrition and assess loan eligibility more effectively. This research underscores the significance of responsible AI practices, ensuring fairness, transparency, and explainability in financial modelling. As machine learning (ML) continues to evolve, deep learning, agentic retrieval-Augmented generation (RAG) and other advanced techniques hold potential for further optimizing predictive performance, greater customer experience and adapt to banking regulations. The proposed model provides valuable insights, enabling banks to refine retention strategies, mitigate customer churn, and enhance lending decisions with greater precision and reliability.

Keywords: Data Analysis, Predictive Modelling, Cross-Sell, Machine Learning (ML), Customer Retention, Credit Decisions, Artificial Intelligence (AI).

1. Introduction

In the rapidly evolving financial landscape, banks and financial institutions must navigate a complex environment characterized by shifting customer behaviors, increasing regulatory requirements, and heightened competition. One of the most pressing challenges in the banking sector is customer churn, which refers to the attrition of existing customers. Customer churn can significantly impact a bank's revenue streams, customer satisfaction, and overall profitability [1]. As digital banking services continue to expand, financial institutions must carefully recognize at-risk customers and implement procedure to enhance customer retention. Additionally, accurately predicting loan eligibility amounts is critical for providing personalized financial solutions while effectively managing credit risk [2]. Traditional methods of assessing loan eligibility often rely on rigid, rule-based approaches that fail to capture the dynamic nature of customer financial behavior. The fusing of Machine Learning (ML) models into banking operations has emerged as a progressive solution to address these challenges, offering enhanced predictive capabilities and improved decision-making processes [3].

ML algorithms enable financial institutions to analyze vast volumes of customer data, extracting meaningful insights that facilitate accurate predictions and risk assessments. By leveraging supervised and unsupervised learning techniques, banks can develop sophisticated models to classify customers based on churn risk, predict loan eligibility with higher precision, and optimize operational strategies [4]. These data-driven approaches empower banks to make informed decisions, leading to better

resource allocation, improve customer delight, and a driven edge in the financial industry [5]. The predictive power of ML algorithms stems from their ability to uncover intricate patterns in customer behavior, transaction history, and financial profiles, which may not be immediately apparent through conventional analysis [6].

One of the fundamental applications of ML in banking is customer churn prediction. Banks can utilize historical customer data, including transaction frequency, engagement metrics, service usage patterns, and demographic details, to build models that classify customers as high or low churn risks [7]. Logistic regression, random forests, decision trees and deep learning models such as neural networks have demonstrated high efficacy in identifying churn-prone customers with remarkable accuracy [8]. By proactively recognizing at-risk customers, banks can implement targeted retention procedures, such as customize offers, magnify customer service, and loyalty programs, to mitigate churn and improve long-term customer relationships [9].

Similarly, ML-driven loan eligibility assessment has revolutionized the credit evaluation process by enhancing predictive accuracy and reducing biases associated with traditional scoring models. Conventional credit assessment methods often rely on standardized credit scores, which may not fully capture an applicant's financial potential or risk profile [10]. ML models, however, can incorporate diverse variables, including employment history, spending behavior, previous loan repayment records, and even alternative credit data sources such as utility bill payments and social media activity [11]. By applying methods such as support vector machines, gradient boosting and ensemble learning, banks can create more precise and

equitable loan eligibility frameworks that optimize risk management while expanding financial inclusion [12].

The integration of ML technologies into the banking sector is not without its challenges. Issues related to data privacy, model clarity, and governmental compliance must be carefully managed to ensure ethical and transparent decision-making processes. Additionally, ML models require continuous monitoring and updating to maintain their effectiveness in dynamic financial environments. However, despite these challenges, the benefits of ML-driven banking solutions far outweigh the limitations, making them indispensable tools for modern financial institutions seeking to enhance efficiency, reduce risk, and deliver superior customer experiences.

2. Literature Review

Jadhav et al. [13] conducted a study on explainable artificial intelligence (XAI) in churn prediction, emphasizing the need for transparency in machine learning models. They explored various interpretability techniques and demonstrated their effectiveness in improving trust and decision-making in business applications. Their study highlighted the advantages of explainable models in financial institutions, enabling stakeholders to understand predictions better.

Muneer et al. [14] proposed ML-based approach for predicting customer churn in the banking industry. They evaluated multiple algorithms, including Support Vector Machine, Random Forest, and Logistic Regression, to determine the best performing model. Their findings indicated that Random Forest achieved the highest reliable prediction values, showcasing its robustness in handling churn prediction tasks. Their work contributes to optimizing customer retention strategies by leveraging advanced data-driven techniques.

Ullah et al. [15] introduced a churn prediction model using Random Forest, analyzing different ML learning techniques for churn forecast in the telecom sector. Their study identified key influencing factors contributing to customer attrition, providing insights into customer retention strategies. They concluded that Random Forest exceeded other models in terms of predictive accuracy and interpretability, making it a viable approach for various domains beyond telecom.

Verbeke et al. [16] focused on developing comprehensible churn prediction models using advanced rule induction techniques. Their research demonstrated that rule-based methods could enhance interpretability while maintaining high predictive performance. The study emphasized the importance of balancing accuracy with model transparency in customer churn prediction, a crucial consideration for regulatory compliance in financial services.

Văduva et al. [17] proposed an improved churn detection model for the banking sector, integrating probability calibration techniques to enhance machine learning model performance. Their work addressed issues related to model confidence and reliability, ultimately improving the decision-making process in financial institutions. The study highlights the importance of probability calibration in reducing false positive rates in churn predictions.

Rajamohamed and Manokaran [18] developed a churn forecast model based on uneven clustering and supervised learning techniques. Their research introduced a novel feature selection mechanism that improved classification accuracy. The study focusses on importance of clustering in refining the feature space for better churn detection, demonstrating that

integrating unsupervised learning techniques with supervised models can enhance prediction effectiveness.

Chopde et al. [19] explored the application of artificial neural networks (ANNs) in churn modeling. They conducted extensive experiments using different ANN architectures, concluding that deep learning-based approaches significantly improved churn prediction accuracy. Their study provided a comparative analysis of ANN versus traditional machine learning methods, emphasizing the potential of deep learning in capturing intricate customer behavior patterns.

Sai et al. [20] investigated customer segmentation in banking for churn prediction, leveraging machine learning techniques. Their research emphasized the role of segmentation in identifying high-risk customers and tailoring retention strategies. They found that clustering methods such as K-Means and hierarchical clustering significantly enhanced prediction accuracy, showcasing how customer segmentation can be effectively used to mitigate attrition rates.

Pathak et al. [21] introduced a hybrid churn prediction framework that combined customer churn analysis with personalized recommendations. Their study aimed to improve customer engagement by integrating predictive analytics with recommendation systems. They demonstrated that personalized retention strategies reduced customer attrition rates effectively, underscoring the significance of a holistic approach to customer relationship management.

Xiahou and Harada [22] examined the application of AdaBoost and BP neural networks for churn prediction in the e-commerce industry. Their study compared the effectiveness of ensemble learning with traditional neural networks, concluding that AdaBoost improved prediction performance by enhancing weak classifiers. Their findings indicate that ensemble methods are particularly effective in handling large-scale customer data with varying degrees of complexity.

Naveen et al. [23] applied fuzzy ARTMAP for churn prediction in bank credit cards, showcasing its capability to handle imbalanced datasets. Their study demonstrated that fuzzy logic-based approaches provided better generalization and adaptability in dynamic financial environments. This work is significant in scenarios where traditional machine learning models struggle with sparse and unevenly distributed data.

Dharwadkar and Patil [24] focused on customer maintenance and credit risk analysis using ANN, SVM, and DNN models. Their research highlighted the advantages of deep learning models in capturing complex patterns and dependencies in customer data. They concluded that hybrid approaches combining different algorithms yielded superior results in churn prediction and credit risk assessment. This study furnishes to the growing body of research that integrates deep learning techniques with conventional machine learning models for more reliable financial forecasting.

3. Materials and Methods

The proposed system utilizes machine learning techniques to enhance customer retention and loan decision-making in the financial sector. It employs a stacking ensemble classifier, integrating multiple models such as Voting Classifier, Extra Trees Classifier, Support Vector Machine (SVM), Random Forest, Decision Tree, and Logistic Regression as the meta-classifier. The system processes two datasets: Churn Data for customer attrition prediction and Credit Eligibility for loan approval assessment. To address data imbalance, RandomOverSampler is applied, ensuring balanced class representation. Feature engineering methods such as one-hot

encoding and standardization are used for data preprocessing. The model undergoes training with five-fold cross-validation, optimizing predictive performance. The framework is designed for real-time deployment, allowing financial institutions to analyze churn risk and loan eligibility through an AI-powered interface. Explainability methods, including SHAP and LIME, enhance model transparency, ensuring fairness and trust. This system offers a data-driven approach to refining retention strategies and loan assessments.

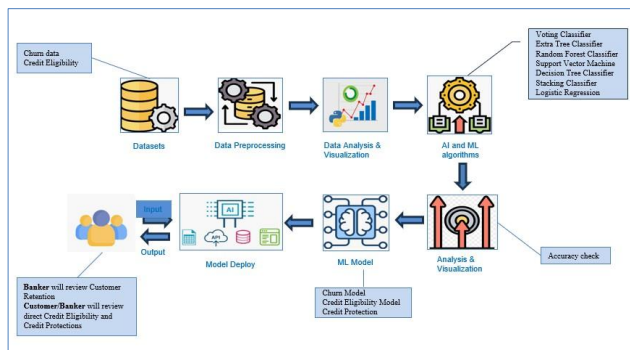


Fig.1 Proposed block diagram

The system architecture represents an AI-powered framework for predicting customer churn and loan eligibility using machine learning. It begins with the collection of two datasets: Churn Data and Credit Eligibility. The data undergoes preprocessing, including cleaning, encoding, and handling class imbalance using RandomOverSampler. Next, data analysis and visualization are performed to identify key patterns. The processed data is then fed into various machine learning algorithms, including Voting Classifier, Extra Trees Classifier for customer retention, Random Forest, SVM, Decision Tree, through Stacking Classifier, and Logistic Regression. The trained models undergo an accuracy check before deployment. Once deployed, financial professionals can access insights for customer retention and loan eligibility decisions. The system ensures transparency through AI-driven analysis, improving decision-making efficiency in the banking sector.

3.1. Dataset Collection:

This study utilizes two datasets: Churn Data and Credit Eligibility Data, which are collected from financial institutions. The Churn Data helps in predicting customer retention by analyzing customer demographics and financial behavior, while the Credit Eligibility Data evaluates loan approval likelihood based on various applicant attributes. These datasets undergo preprocessing to handle missing values, normalize data, and balance class distributions before applying machine learning models.

i) Churn Data:

The Churn Data consists of various customer attributes relevant to predicting churn behavior. It includes Customer ID and Name for identification. Credit Score represents the customer's financial reliability. The Branch indicates which bank branch the customer belongs to. Gender is categorized as Male or Female. Age is recorded as numerical values for customers aged 18 and above. Tenure constitutes the number of years the customer has been with the bank. Balance is stored as a numerical value. NumOfProducts indicates the number of bank products used by the customer. Has Credit Card and Is Active Member are boolean attributes

reflecting the customer's engagement. Estimated Salary is a numerical attribute, and Exited (boolean) indicates if the customer has left the bank.

ii) Credit Eligibility Data:

The Credit Eligibility Data includes multiple features used to assess an applicant's loan eligibility. Loan_ID uniquely identifies each loan application. Gender is classified as Male or Female. Married is a boolean value indicating marital status. Dependents represent the number of dependents financially reliant on the applicant. Education classifies applicants as Graduate or Not Graduate. Self_Employed is a boolean attribute indicating self-employment status. ApplicantIncome and CoapplicantIncome are numerical values representing financial earnings. LoanAmount denotes the loan amount requested. Loan_Amount_Term specifies the repayment term in months. Credit_History indicates past repayment behavior, and Property_Area categorizes applicants based on Urban, Semi-Urban, or Rural locations.

3.2. Pre-Processing:

Pre-processing transforms raw data into a clean, structured format by handling missing values, encoding categorical variables, balancing data, and splitting datasets.

Feature Selection: Feature selection is an essential step to remove irrelevant or redundant data. In this dataset, Loan_ID and Credit_History are dropped as they do not provide meaningful contributions to predicting credit eligibility. Loan_ID is a unique identifier with no impact on the classification, while Credit_History may introduce bias. Removing unnecessary columns reduces dimensionality, simplifies the model, and improves computational efficiency. This step ensures that only relevant features are used, leading to better generalization and improved model performance during training and testing.

Handling Missing Values: Missing values can negatively impact machine learning models by introducing inconsistencies and bias. In this dataset, missing values are handled by removing incomplete rows to ensure data integrity. Although imputation techniques could be used, dropping missing values is a straightforward approach when data loss is minimal. This ensures that the dataset remains clean and reliable. By eliminating incomplete records, the risk of errors during model training is minimized, leading to better accuracy and more meaningful predictions in credit eligibility assessment.

Encoding Categorical Variables: Categorical variables need to be converted into numerical format for ML models. Here, Label Encoding is applied to features like Gender, Married, Education, Self_Employed, Property_Area, Loan_Status, and Dependents, assigning each category a unique integer value. This conversion allows algorithms like Decision Tree, Random Forest, and SVM to process the data effectively. Label encoding preserves categorical relationships without increasing dimensionality, making the dataset suitable for model training. Proper encoding ensures that the model correctly interprets categorical attributes, leading to accurate credit eligibility predictions.

Handling Imbalanced Data: Class imbalance can cause biased predictions, where the model favors the majority class. To address this, Random Over-Sampling (ROS) is applied, increasing the minority class instances until both classes have an equal count. Initially, the dataset had Loan_Status: 1 → 367, 0 → 156, which could lead to poor classification of loan denials. After applying ROS, both classes contain 367 instances, ensuring that the model learns to distinguish both outcomes fairly. This improves the classifier's performance, particularly in identifying applicants who are not eligible for loans.

3.3. Training and Testing

For effective evaluation of the model, the dataset is split into training (90%) and testing (10%) sets with stratify=y to ensure the models are well-equipped to handle all categories, leading to more robust and reliable outcomes. This division ensures that the model learns patterns from a large portion of the data while leaving unseen examples for testing. This approach prevents overfitting and ensures that the model hypothesizes well to new data, ultimately improving its trustworthiness in predicting loan eligibility in real-world applications.

3.4. Algorithms:

Voting Classifier: A Voting classifier is an ensemble learning technique used in machine learning for combining the predictions of multiple individual classifiers to make a final decision. It is a type of "majority voting" approach where each classifier in the ensemble contributes to the final prediction, and the class with the most votes is chosen as the final output. The voting classifier can be used for both classification and regression tasks, but it is most commonly employed for classification. In a typical scenario, the voting classifier consists of several base classifiers, which can be of different types or trained using different algorithms. These base classifiers may include decision trees, support vector machines, logistic regression, k-nearest neighbors, or any other suitable classification model. During the prediction phase, each base classifier makes its own individual prediction based on the input data. For a classification task, each classifier assigns a class label to the data point. In the case of regression, each classifier provides a numeric prediction. Then, the voting classifier combines these individual predictions using one of two methods: hard voting or soft voting.

Extra Tree Classifier: The Extra Trees Classifier, short for Extremely Randomized Trees Classifier, is a popular machine learning algorithm used for classification tasks. It belongs to the ensemble learning family, which combines multiple base models to make more accurate predictions. Extra Trees is an extension of the Random Forest algorithm and shares some similarities with it.

What sets the Extra Trees Classifier apart is its extreme randomness during the tree-building process. In a typical decision tree or Random Forest, the algorithm selects the best feature to split the data at each node based on some criterion like Gini impurity or entropy. However, in Extra Trees, the splitting feature is chosen entirely at random, without evaluating their significance thoroughly. This randomness helps reduce overfitting and improves computational efficiency.

To build an Extra Trees Classifier, multiple decision trees are created using bootstrapped samples from the training data, just like in Random Forests. However, each tree is further randomized by selecting random subsets of features at each node for determining the splits. This makes Extra Trees less sensitive to noisy features and more robust to outliers.

During the prediction phase, the results from all individual trees are aggregated. In classification tasks, the class with the majority of votes from the trees is selected as the final prediction.

Extra Trees can be a powerful tool for classification tasks, especially when you have a large dataset with many features. Its ability to handle noisy data and reduce overfitting makes it a valuable addition to the machine learning practitioner's toolkit. However, due to its inherent randomness, it might not provide easily interpretable insights into the importance of features compared to traditional decision trees. Nonetheless, it is a valuable

ensemble method for improving classification accuracy and handling complex datasets.

Random Forest Classifier: Random Forest is an ensemble learning procedure that constructs multiple decision trees to enhance classification accuracy and lower overfitting. It aggregates predictions from individual trees to produce a robust final decision. This algorithm enhances model generalization, ensuring stable performance even with noisy data. By leveraging feature importance, it aids in identifying key factors influencing predictions. Its resilience to outliers and capability to handle both numerical and categorical variables make it a suitable choice. Additionally, Random Forest excels in high-dimensional datasets, offering improved interpretability and efficiency while reducing variance compared to single decision trees.

Support Vector Machine (SVM): Support Vector Machine is a supervised learning model that finds the perfect hyperplane separating different classes. It expands the limits between data points from different categories, ensuring better generalization. Kernel functions enhance its capability to classify non-linearly separable data. With strong mathematical foundations, SVM provides high accuracy in complex classification problems. Regularization parameters help balance bias and variance, preventing overfitting. The model works well for small-to-medium-sized datasets where feature spaces are well-defined. Its effectiveness in handling high-dimensional spaces and preventing misclassification makes it a reliable choice for predictive analytics.

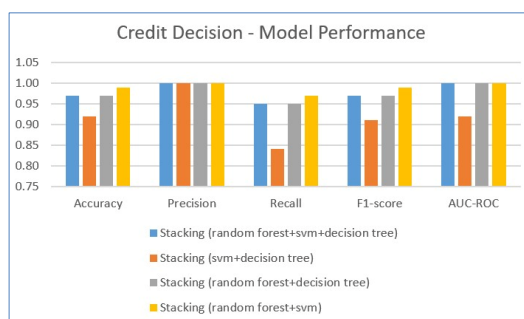
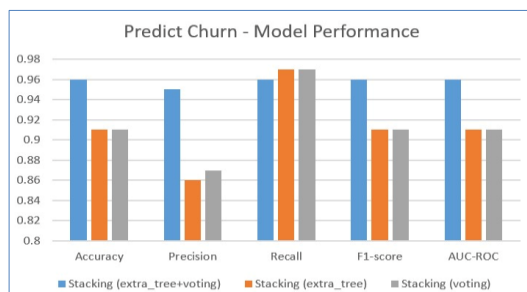
Decision Tree Classifier: This Classifier is a rule-based model that partitions data into smaller subdivisions through a hierarchical structure. It follows a flowchart-like approach where each node represents a decision based on feature values. Simplicity and interpretability make it effective for understanding relationships between variables. However, single trees tend to overfit, making pruning techniques necessary to enhance generalization. This algorithm efficiently handles missing values and categorical variables, adapting well to varied datasets. By breaking down complex decisions into sequential steps, it provides transparency while maintaining computational efficiency, ensuring clear, structured, and data-driven classification outcomes.

Stacking Classifier: Stacking Classifier integrates multiple base models to increase predictive performance by leveraging their strengths. It integrates predictions from SVM, Random Forest and Decision Tree classifiers, using Logistic Regression as a meta-model for final decision-making. By learning patterns from different classifiers, it improves accuracy, reduces individual model weaknesses, and enhances robustness. Cross-validation ensures optimal feature extraction, minimizing overfitting. Stacking allows for a more generalized model by capturing diverse insights from multiple algorithms. This hierarchical learning approach ensures better decision-making through collective intelligence, ultimately boosting overall predictive accuracy and reliability.

Logistic Regression: Logistic Regression serves as the final estimator in the stacking model, consolidating predictions from base classifiers. It applies the logistic function to predict probabilities, ensuring outputs fall between 0 and 1. The model is efficient for binary classification, distinguishing between two possible outcomes. Regularization techniques enhance stability by preventing overfitting. Its interpretability and simplicity make it useful for understanding feature influence. Logistic Regression excels in scenarios where linear decision boundaries exist, contributing to the final stacked model's efficiency. It effectively refines combined predictions, making it a crucial component for robust and accurate classification.

4. Experimental Results

Fig.1 Model Performance Comparison



** 97% accuracy is ideal to avoid overfit and errorless in predictions

Fig.2 Classification Report Heatmap for stacking 97%

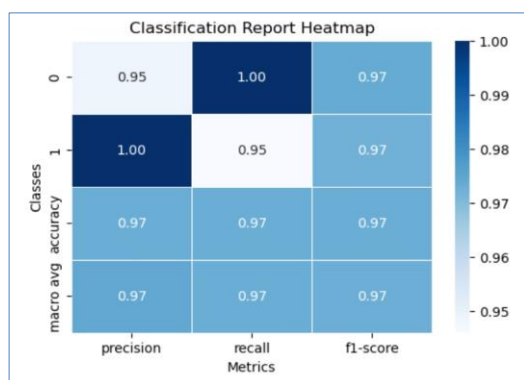


Fig.3 ROC-AUC Curve for stacking 97% 's accuracy

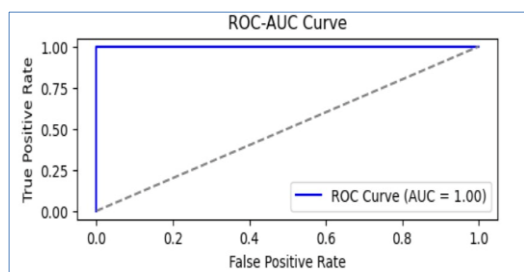
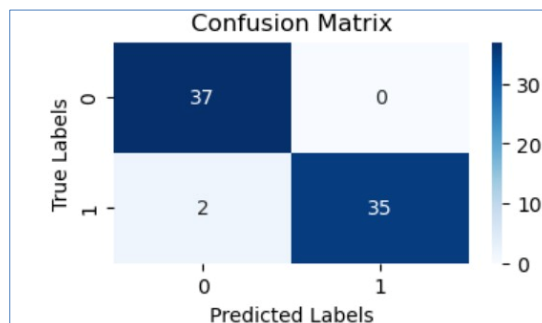


Fig.4 Confusion Matrix for Model Accuracy



Key Observations: Overall Stacking (random forest+svm) was resulted maximum accuracy but in order to keep the model performance ideal and to avoid overfit and errorless in predictions. Stacking (random forest+svm+decision tree) classifiers has resulted good model performance for $cv=5$, $n_estimators=100$, $random_state=42$. If the dataset contains millions of transactions, we have to further optimize cv , $random_state$ along with correct proposition of train and test to achieve the higher accuracy rate.

5. Conclusion

The analytical process began with data cleaning and preprocessing, ensuring consistency and quality in the Churn Prediction Dataset and Loan Eligibility Dataset. Feature encoding transformed categorical attributes into numerical values, enhancing model performance. Oversampling techniques balanced the dataset, preventing biases in classification. A stacking ensemble model, incorporating Voting and Extra Tree classifier to predict bank churn and Random Forest, Support Vector Machine, and Decision Tree classifiers, was trained to predict loan eligibility. Logistic Regression acted as the meta-classifier, refining the final predictions. The best accuracy achieved on the public test set was optimized through hyperparameter tuning and cross-validation, ensuring reliable predictions. This application provides valuable insights for banks in identifying potential customer churn and determining loan eligibility with higher accuracy. The integration of advanced machine learning techniques enhances financial decision-making, minimizing risks and improving customer retention strategies. The Churn Prediction Dataset and Loan Eligibility Dataset played a crucial role in training the model effectively, making this a robust and efficient predictive system.

The future scope of this system includes enhancing real-time automation for detecting bank churn and loan eligibility using advanced AI techniques. Integrating the model with agentic retrieval-Augmented generation (RAG) systems can help in quick decision-making, improve customer experience, reducing manual efforts and real time validation of customer profile with heterogenous data sources and adaptive to banking regulations. The system can be deployed as a web or desktop application, allowing seamless access for banks and financial institutions. Future improvements include incorporating deep learning algorithms for better prediction accuracy and expanding the dataset to include more diverse customer attributes. Additionally, real-time data streaming and cloud-based deployment can enhance scalability and performance. With continuous optimization, this system can evolve into a comprehensive financial risk assessment tool in an AI-driven environment.

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